University of Engineering & Technology Peshawar

Programming Fundamentals (C++) Lab

LAB REPORT 2

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COUSRE:	**EE-170LCOMPUTER PROGRAMMING LAB**
DATE:	**3/6/26**
LAB TITLE:	**C++ DATA TYPE,MEMORY SIZE AND MEMORY ADDRESSS**

SUBMITTED TO ENGR RIZWAN ULLAH

Learning Objectives:

The objective of this lab is:

- To understand different data types in C++
- To learn how to declare variables

- To observe the memory size of different data types
- To find the memory addresses of variables using the address operator (&)

Theory:

In C++, variables are used to store data in memory. Each variable has:

- Data Type – defines what kind of data it can store.
- Memory Size – the amount of memory allocated for the variable.
- Memory Address – the location where the variable is stored in RAM.

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Data Type	Description
int	Stores integer numbers
float	Stores decimal numbers
double	Stores large decimal numbers
char	Stores a single character
bool	Stores true or false

Algorithm:

1. Start the program
2. Declare variables of different data types
3. Assign values to variables
4. Print the values of variables
5. Use sizeof() to print memory size
6. Use & operator to print memory addresses
7. End the program

Program code:

```
#include <iostream>
using namespace std;
int main() {
   int a = 10;
   float b = 5.5;
   double c = 20.99;
   char d = 'A';
   bool e = true;

   cout << "VALUES:\n";
   cout << "int a = " << a << endl;
```

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```
    cout << "float b = " << b << endl;
    cout << "double c = " << c << endl;
    cout << "char d = " << d << endl;
    cout << "bool e = " << e << endl;

    cout << "\nMEMORY SIZE (in bytes):\n";
    cout << "Size of int: " << sizeof(a) << endl;
    cout << "Size of float: " << sizeof(b) << endl;
    cout << "Size of double: " << sizeof(c) << endl;
    cout << "Size of char: " << sizeof(d) << endl;
    cout << "Size of bool: " << sizeof(e) << endl;

    cout << "\nMEMORY ADDRESS (Location):\n";
    cout << "Address of a: " << &a << endl;
    cout << "Address of b: " << &b << endl;
    cout << "Address of c: " << &c << endl;
    cout << "Address of d: " << (void*)&d << endl;
    cout << "Address of e: " << &e << endl;
    return 0;
}
```

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Program Output

VALUES:

int a = 10

float b = 5.5

double c = 20.99

char d = A

bool e = 1

MEMORY SIZE (in bytes):

Size of int: 4

Size of float: 4

Size of double: 8

Size of char: 1

Size of bool: 1

MEMORY ADDRESS (Location):

Address of a: 0x61ff08

Address of b: 0x61ff04

Address of c: 0x61fef8

Address of d: 0x61fef7

Address of e: 0x61fef6

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RESULT:

The program successfully displayed;

- The program successfully displayed:
- Values of different variables
- Memory size of each data type
- Memory addresses of variables

Conclusion:

From thos experiment we learned;

- Different **C++ data types store different kinds of data**
- Each data type occupies **different memory sizes**
- The **address operator (&)** can be used to find the memory location of variables

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